

Statistical Analysis Report: Student Performance Data

Learning Intervention Study in GP1 vs GP2

Executive Summary

This report presents a comprehensive statistical analysis of student performance data from a learning intervention study involving 182 students across two groups: GP1 (n = 84) and GP2 (n = 98). The analysis covers descriptive statistics, independent-samples t-tests for group comparisons, paired t-tests for pre-to-post intervention changes, Pearson correlation analysis, and outlier detection. Key findings show that both groups achieved highly significant improvements in test scores and enjoyment following the intervention, with no statistically significant differences between the two groups on any measured outcome.

1. Dataset Overview

The cleaned dataset contains **182 student records** with **243 variables** and **zero missing values**, confirming 100% data completeness. The core outcome variables analyzed are:

- **Total Pre Test** — Knowledge score before intervention (max 70)
- **Total Post Test** — Knowledge score after intervention (max 70)
- **Test Knowledge Gain** — Post minus Pre-test score
- **Total Pre Enjoyment** — Enjoyment score before intervention
- **Total Post Enjoyment** — Enjoyment score after intervention
- **Total Passion** — Passion for learning scale score
- **Passion Learning Efficiency** — Ratio of passion to learning outcome (0–1)

Attribute	Value
Total Students	182
GP1	84 students
GP2	98 students
Total Variables	243
Missing Values	0 (100% complete)
Pre-Test Questions	70 items
Post-Test Questions	70 items
Enjoyment Items	32 items (Pre + Post)
Passion Items	30 items

2. Descriptive Statistics

The table below presents the mean and standard deviation for each core metric, broken down by group and overall.

Metric	GP1 Mean	GP1 SD	GP2 Mean	GP2 SD	All Students Mean	All Students SD
Total Pre Test	27.98	4.12	27.44	4.61	27.69	4.39
Total_Post_Test	55.90	3.58	55.55	3.21	55.71	3.38
Test_Knowledge_Gain	27.93	5.76	28.11	5.58	28.03	5.65
Total_Pre_Enjoyment	77.11	4.40	77.09	4.35	77.10	4.36
Total_Post_Enjoyment	82.71	3.94	83.16	5.14	82.96	4.62
Total_Passion	75.23	4.49	75.84	4.69	75.55	4.60
Passion_Learning_Efficiency	0.746	0.067	0.735	0.062	0.740	0.064

Key observations:

- Average knowledge gain across all students was **+28.03 points** (from 27.69 to 55.71)
- Enjoyment increased from a pre-mean of **77.10** to a post-mean of **82.96** (+5.86 points)

- Passion Learning Efficiency averaged **0.74**, indicating students converted approximately 74% of their passion into measurable learning gains
- Both groups show nearly identical means on all metrics, suggesting comparable baseline characteristics

3. Group Comparison by Independent Samples t-Test

An independent-samples t-test was conducted to compare GP1 and GP2 on each core metric. The null hypothesis for each test was that the two groups have equal population means.

Metric	GP1 Mean (95% CI)	GP2 Mean (95% CI)	Diff	t	p-value	Sig	Cohen's d	Effect Size
Total Pre Test	27.98 [27.08, 28.87]	27.44 [26.51, 28.36]	+0.54	0.823	0.4116	ns	0.122	Negligible
Total Post Test	55.90 [55.13, 56.68]	55.55 [54.91, 56.19]	+0.35	0.703	0.4828	ns	0.105	Negligible
Test Knowledge Gain	27.93 [26.68, 29.18]	28.11 [26.99, 29.23]	-0.18	-0.218	0.8275	ns	-0.032	Negligible
Total Pre Enjoyment	77.11 [76.15, 78.06]	77.09 [76.22, 77.96]	+0.02	0.024	0.9813	ns	0.004	Negligible
Total Post Enjoyment	82.71 [81.86, 83.57]	83.16 [82.13, 84.19]	-0.45	-0.653	0.5146	ns	-0.097	Negligible
Total Passion	75.23 [74.25, 76.20]	75.84 [74.90, 76.78]	-0.61	-0.893	0.3733	ns	-0.133	Negligible
Passion Learning Efficiency	0.746 [0.73, 0.76]	0.735 [0.72, 0.75]	+0.011	1.110	0.2685	ns	0.165	Negligible

Significance *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$ ns = not significant ($p \geq 0.05$)

Interpretation

None of the seven comparisons reached statistical significance. All p-values are well above the 0.05 threshold (range: 0.27 to 0.98), and all Cohen's d values are below 0.20, indicating negligible effect sizes. This means:

- The two groups were **equivalent at baseline** (pre-test and pre-enjoyment)
- Both groups achieved **equivalent outcomes** after the intervention (post-test, knowledge gain, enjoyment, passion)
- The intervention appears to be **equally effective** regardless of whether students were in the standard or advanced group

4. Pre vs Post Intervention — Paired Samples t-Test

To assess whether the intervention produced significant within-group improvements, paired-samples t-tests were conducted comparing pre- and post-intervention scores for both test knowledge and enjoyment.

Group	Measure	Pre Mean	Post Mean	Gain	t-statistic	p-value	Sig	Cohen's d
GP1	Test Score	27.98	55.90	+27.93	-44.46	< 0.001	***	7.24
GP1	Enjoyment	77.11	82.71	+5.61	-8.28	< 0.001	***	1.34
GP2	Test Score	27.44	55.55	+28.11	-49.91	< 0.001	***	7.08
GP2	Enjoyment	77.09	83.16	+6.07	-9.13	< 0.001	***	1.28
All Students	Test Score	27.69	55.71	+28.03	-66.98	< 0.001	***	7.16
All Students	Enjoyment	77.10	82.96	+5.86	-12.35	< 0.001	***	1.30

Interpretation

All six paired t-tests yielded ($p < 0.001$), indicating highly significant improvements in both test scores and enjoyment following the intervention:

- **Knowledge gain** produced extraordinarily large effect sizes (Cohen's $d \approx 7.1$ – 7.2), meaning the post-test scores were approximately 7 standard deviations higher than pre-test scores — a massive, practically significant improvement
- **Enjoyment gain** also showed large effect sizes (Cohen's $d \approx 1.3$), exceeding the conventional threshold of 0.80 for a large effect
- Both groups show essentially identical patterns of improvement, confirming the intervention was uniformly effective

5. Pearson Correlation Analysis

Pearson correlation coefficients were computed between all pairs of core metrics ($n = 182$) to examine relationships between variables. Only the 10 strongest correlations are presented below; the full 7×7 matrix is available in the Excel workbook.

Variable 1	Variable 2	r	p-value	Sig	Interpretation
Test Knowledge Gain	Total Pre Test	−0.802	< 0.001	***	Strong (negative)
Passion Learning Efficiency	Total Post Test	+0.710	< 0.001	***	Strong (positive)
Passion Learning Efficiency	Total Passion	−0.706	< 0.001	***	Strong (negative)
Test Knowledge Gain	Total Post Test	+0.630	< 0.001	***	Moderate (positive)
Passion Learning Efficiency	Test Knowledge Gain	+0.485	< 0.001	***	Moderate (positive)
Test Knowledge Gain	Total Post Enjoyment	+0.224	0.002	**	Weak (positive)
Total Post Enjoyment	Total Pre Test	−0.179	0.016	*	Negligible (negative)
Total Post Enjoyment	Total Post Test	+0.142	0.057	ns	Negligible
Passion Learning Efficiency	Total Post Enjoyment	+0.121	0.103	ns	Negligible
Test Knowledge Gain	Total Pre Enjoyment	+0.079	0.290	ns	Negligible

Key Findings

1. Pre-Test score strongly predicts Knowledge Gain ($r = -0.802$, $p < 0.001$):

Students who scored lower on the pre-test achieved greater knowledge gains. This ceiling/floor effect is expected — students with lower initial knowledge have more room to improve.

2. Passion Learning Efficiency correlates strongly with Post-Test ($r = +0.710$, $p < 0.001$):

Students with higher learning efficiency relative to their passion achieve better final test scores, suggesting that motivation quality (not just quantity) matters for learning outcomes.

3. Passion Learning Efficiency correlates negatively with Total Passion ($r = -0.706$, $p < 0.001$):

Students with very high passion scores paradoxically show lower efficiency ratios, which may indicate that highly passionate students set higher internal standards or engage in broader exploration beyond what the test measures.

4. Enjoyment and Knowledge Gain show only weak correlation ($r = +0.224$, $p = 0.002$):

While enjoyment is positively associated with learning, the relationship is weak, implying that enjoyment alone is not a strong predictor of knowledge acquisition in this sample.

6. Outlier Detection

Outliers were identified using Z-score criterion: any observation more than 2.0 standard deviations from the group mean was flagged. A total of **58 outlier observations** were detected across 182 students and 7 metrics.

Metric	Outlier Count
Total Pre Test	8
Total Post Test	6
Test Knowledge Gain	7
Total Pre Enjoyment	9
Total Post Enjoyment	10
Total Passion	11
Passion Learning Efficiency	7
Total	58

Interpretation

The outlier rate is approximately **4.5% per metric** ($58 \div (182 \times 7) \approx 4.6\%$), which is within the expected statistical range for a normally distributed dataset (approximately 4.6% of observations fall beyond ± 2 SD by definition). No single metric shows a disproportionately high outlier rate. **No outlier records were removed** from the main analysis, as their presence is statistically expected and they do not distort group-level findings.

7. Summary of Findings

Research Question	Finding	Statistical Support
Did the intervention improve knowledge?	Yes — +28 points average gain	Paired t-test: $t = -66.98$, $p < 0.001$, $d = 7.16$
Did the intervention improve enjoyment?	Yes — +5.86 points average gain	Paired t-test: $t = -12.35$, $p < 0.001$, $d = 1.30$
Is there a difference between GP1 and GP2?	No — groups are statistically equivalent	All p-values ns (0.27–0.98), all $d < 0.20$
Does passion predict learning efficiency?	Yes — moderate positive relationship	$r = +0.485$, $p < 0.001$
Does pre-test score predict gain?	Yes — strongly (inverse relationship)	$r = -0.802$, $p < 0.001$
Does enjoyment predict knowledge gain?	Weakly — small positive relationship	$r = +0.224$, $p = 0.002$
Are data quality issues present?	No — 100% complete, normal outlier rate	Quality sheet: all columns 100% complete

8. Recommendations

Based on the analysis, the following actions are recommended:

1. **Report equivalence of groups as a strength** the near-identical outcomes across GP1 and GP2 suggest the intervention design is robust and reproducible across different student cohorts.
2. **Highlight the exceptional knowledge gain effect size** Cohen's $d \approx 7.16$ for test score improvement is extraordinarily high, substantially exceeding the conventional "large effect" threshold of 0.80, and strongly demonstrates the intervention's academic impact.
3. **Investigate the passion-efficiency paradox** the negative correlation ($r = -0.706$) between Total Passion and Passion Learning Efficiency warrants qualitative exploration; highly motivated students may need additional challenge or differentiated tasks.
4. **Monitor the 58 flagged outliers** while statistically expected, reviewing extreme cases ($Z > 3.0$) may reveal individual students who require additional academic support or who represent data entry errors.
5. **Address the weak enjoyment-knowledge link** since enjoyment does not strongly predict knowledge gain ($r = 0.224$), future research should explore additional mediating variables such as self-efficacy, metacognition, or prior experience.